

How to build a Claims MCP agent

(the Smart Claims story — explained for everyone ✨)

- ① customer chats →
- ② agent calls the MCP tool →
- ③ case is created in Pega →
- ✓ claim logged 🎉

no manual
data entry
just chat 💬

swipe →

What is it?

A **Claims Assistant** — a chat agent that lets a customer report a motor accident and have the claim logged automatically.

Think of **MCP as a USB-C port for AI**:
one plug connects the agent to Pega, email, maps — any tool.



Without MCP

A human agent types every claim into Pega by hand.



With MCP

The agent calls a tool and creates the case in seconds.

What you'll build

An MCP tool called **pega_create_case** that the agent calls from chat.

● ● ● claims chat

I had an accident, policy POL123456710

🔧 pega_create_case(...)



→ Case ID: FAI-SMARTCLAIM S-145003

ALONG THE WAY YOU'LL LEARN:

- How to expose a function as an MCP tool
- How to host it on Azure Container Apps
- How to wire it into Copilot Studio

Before you start

YOU'LL NEED:



Python 3.12

The MCP server runs on FastMCP



An Azure account

To host the server on Container Apps



Pega + Copilot Studio

Where cases live and the agent runs

THEN INSTALL THE SDK:

```
$ pip install fastmcp httpx
```

Create the server

Spin up an empty MCP server with **FastMCP**.

```
● ● ● main.py
```

```
from mcp.server.fastmcp import FastMCP
```

```
# name it anything you like
```

```
mcp = FastMCP("pega-claims-mcp")
```



FastMCP handles all the protocol plumbing — you only write the tools.

Add the case tool

The `@mcp.tool()` decorator turns a function into something the agent can call.

● ● ● PegaCreateCase.py

```
@mcp.tool(name="pega_create_case")
async def pega_create_case(
    PolicyNumber: str, EmailID: str,
    case_type_id: str):
    """Create a Smart Claim case in Pega.
    Use when a customer reports a claim."""
    return await client.create_case(...)
```

💡 The **docstring is critical** — that's how the agent decides when to call your tool.

Test it locally

Before connecting Copilot, sanity-check the tool from a local MCP client.

```
$ python -m app.main
```

TOOL: PEGA_CREATE_CASE

PolicyNumber: POL123456710

OUTPUT

Case ID: FAI-SMARTCLAIM S-145003

Catches bugs before the agent does.

Deploy to Azure

Package as a container and push to **Azure Container Apps**.

terminal

```
# build & push the image
$ docker build -t claims-mcp .
$ az containerapp up \
  --name pega-mcp --ingress external
```

⚠ Set the env vars — PEGA_CLIENT_ID, PEGA_CLIENT_SECRET, the token URL and the API base. The MCP server reads them at runtime.

Wire into Copilot

In Copilot Studio → Tools → Add an MCP server, point it at your Azure URL.

Server URL

<https://pega-mcp.azurecontainerapps.io/mcp>

● pega-claims-mcp

✓ Connected



Tools:

pega_create_case



The agent can now create cases!

Stuck? Quick fixes



"Authentication not configured"

Connect the MCP server in Copilot's connection manager, then retry.



Tool fires too early

Add a gate in instructions — validate policy before any tool call.



Empty "Error executing tool"

Check the API base URL and that env vars match your working setup.



Wrong case type id

Hardcode it in the tool config so the agent can't guess it.

Still stuck? Turn on **DEBUG logging** in Azure — the log stream prints the real Pega error.

YOU DID IT

Ship it. Demo it. Scale it.

You now know how a single MCP tool
turns a chat into a real Pega case — with
full control.

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